

SHAJEESH K U

Senior AI Engineer | Computer Vision | Generative AI

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PROFESSIONAL SUMMARY

Senior AI Engineer with 13+ years of experience in computer vision, deep learning, and production-grade Generative AI. At Detect Technologies, architected and deployed real-time multi-camera video analytics and VLM-based inference pipelines across global industrial sites (Shell, IOCL, BPCL, Subsea7, Bechtel), achieving a **95% reduction in workplace accidents** and **90% reduction in manual alert review workload**. Progressed from CV Intern to Associate Manager AI in 4 years. Research background with **peer-reviewed publications** in IEEE and Elsevier journals.

PROFESSIONAL EXPERIENCE

Detect Technologies Chennai, Tamil Nadu | Mar 2022 – Present

CV Intern → AI Software Engineer → Senior AI Engineer → Associate Manager AI

Lead a team of 4–5 AI engineers and 2–3 interns building production computer vision and GenAI systems for industrial workplace safety.

Architected the **Intelligent Video Analytics Framework** — a scalable multi-stream CV system for PPE violation detection, intrusion monitoring, ANPR, vehicle speed estimation, confined space headcount, and emissions analysis; deployed across global sites for Shell, IOCL, BPCL, Subsea7, NRL, Vedanta BALCO, Valaris, and ADES KSA.

Designed the **GenAI Inference Engine for Observation Validation**: a hybrid VLM pipeline (Qwen3, InternVL3.5, Molmo, LLaVA-NeXT, SAIL-VL, PaddleOCR-VL) with dual-mode deployment (on-premise FastAPI + AWS Bedrock) that reduced manual alert review by **90%** and near-eliminated false positives before client notification. Led onsite deployment at **ADES, Saudi Arabia**, delivering the AI Safety Assistant end-to-end and contributing to a **70% revenue increase** for the AI division.

Engineered edge AI deployments on NVIDIA Jetson Nano, ARM Cortex, and Raspberry Pi for bandwidth-constrained industrial environments.

Delivered **95% reduction in workplace accidents** across client sites through real-time PPE compliance and safety monitoring at scale.

MES College of Engineering Kuttippuram, Kerala | Jan 2013 – May 2022

Assistant Professor, Computer Science & Engineering

Cultivated expertise in Machine Learning, Computer Vision, and Python for 9 years, bridging the gap between theoretical AI concepts and applied engineering for undergraduate and postgraduate students.

Spearheaded advanced AI research initiatives, supervising M.Tech projects in computer vision; successfully guided research through to publication in high-impact IEEE and Elsevier journals.

Empanelled Trainer for ICT Academy of Kerala: Designed and executed industry-aligned curricula in Full-Stack development and Applied AI, equipping engineering graduates with production-ready technical skills.

Mentored student-led technology startups, providing technical architectural guidance for successful implementation of applied AI and software development solutions.

Amrita School of Engineering Coimbatore, Tamil Nadu | Oct 2012 – Nov 2012

Research Project Staff — Centre for Excellence in Computational Engineering and Networking

Spearheaded the development of novel speech signal denoising algorithms leveraging compressive sensing and Savitzky–Golay filters; provided technical leadership for the project resulting in **3 peer-reviewed international publications** detailing signal processing and AI research findings. Provided technical leadership in co-organizing national-level workshops on Sparse Image & Signal Processing at Amrita University Coimbatore.

KEY PROJECTS

GenAI Inference Engine for Observation Validation — Detect Technologies

Developed a production-grade hybrid VLM system at Detect Technologies to automate the validation of safety alerts from primary CV models, effectively **eliminating false positives** prior to client notification. Implemented a dual-mode architecture utilizing both on-premise FastAPI model servers and AWS Bedrock cloud inference.

Models: Qwen3, InternVL3.5, Molmo, LLaVA-NeXT, SAIL-VL, PaddleOCR-VL

Architecture: On-premise FastAPI + AWS Bedrock (dual-mode deployment)

✓ **Outcome**: 90% reduction in manual alert review; near-elimination of false positives across all production client sites

Intelligent Video Analytics Framework — Detect Technologies

Architected a scalable multi-stream computer vision framework designed for real-time monitoring of industrial environments. Deployed globally for major clients including Shell, BPCL, and Bechtel to ensure comprehensive workplace safety compliance.

Modules: PPE detection, red zone intrusion, ANPR, vehicle speed monitoring, confined space tracking, flare/smoke emissions analysis
✓ Outcome: 95% reduction in workplace accidents; 70% improvement in operational efficiency

ClearSight-AI Safety Manager — Detect Technologies

Engineered an AWS Rekognition-powered platform for worker authentication and monitoring in confined spaces. Integrated real-time IoT sensor data for environmental thresholds with automated safety reporting protocols.

✓ Outcome: 95% reduction in unauthorized confined space entries and related incidents

Automated Prompt Optimizer for VLMs — Detect Technologies

Developed a production-grade iterative tuning pipeline for Vision-Language Models at Detect Technologies to monitor industrial safety compliance. Engineered a self-corrective evaluation framework that autonomously refines model prompts by isolating classification failures through ground-truth validation, effectively removing manual intervention from the optimization cycle.

Models/Tools: Qwen3-VL, InternVL3.5, Molmo, Zero-Shot Classification, Confusion Matrix Metrics (FP/FN)
✓ Outcome: 50% reduction in prompt engineering efforts; 80% acceleration in global use-case deployment timelines

Optimized Video Transmission Suite — Detect Technologies

Engineered a high-availability video relay infrastructure optimized for bandwidth-constrained industrial environments. Developed Staggered Relay for asynchronous chunked data transmissions and Video Relay Live to ensure secure, real-time remote monitoring during active AI inference cycles.
✓ Outcome: Facilitated successful integration within highly restricted network environments where 90% of rival solutions failed to preserve operational connectivity.

Automated Active Learning & Model Evolution Framework — Detect Technologies

Developed a self-evolving active learning pipeline designed to dynamically calibrate production Vision Transformer (ViT) models to diverse site-specific client domains. Orchestrated a robust edge-to-cloud telemetry infrastructure that automates batch uploads to AWS S3 when inference confidence falls below a 50% threshold. Engineered an agentic AI-driven auto-labeling workflow using Gemini API, integrated with a Human-in-the-Loop (HITL) interface for accelerated dataset curation and parameter-efficient refinement of classification heads.
Tech Stack: Vision Transformers (ViT), Agentic AI, Gemini API, AWS S3, PEFT (LoRA), HITL Web Interfaces.
✓ Outcome: Increased model accuracy from 80% to 95%+ across global environments, significantly enhancing field reliability and client satisfaction.

EDUCATION

M.Tech in Computational Engineering and Networking | Amrita Vishwa Vidyapeetham University, Coimbatore, Tamil Nadu | 2010 – 2012
| CGPA: 8.29
B.Tech in Computer Science and Engineering | University of Calicut, Sahrdaya College of Engineering and Technology, Thrissur | 2005 – 2009 | Percentage: 64.6

TECHNICAL SKILLS

AI, Deep Learning & Computer Vision	Generative AI & Multimodal Systems
• Expertise: Object Detection & Tracking, SAM 3, Human Pose Estimation, OCR, Anomaly Detection, ANPR • Libraries & Frameworks: YOLO (v5–26), OpenCV, PaddleOCR, PyTorch, TensorFlow/Keras, Scikit-learn	• Vision-Language Models (VLMs): Qwen3-7B-Instruct, InternVL3-8B, Molmo-7B, LLaVA-NeXT, MiniCPM, Phi, Ovis1.6 • LLM Orchestration: LangChain, LlamaIndex, OpenAI SDK, AWS Bedrock • Methodologies: RAG, LoRA, QLoRA, Advanced Prompt Engineering
Infrastructure, DevOps & Edge Computing	Systems, Languages & Agentic Dev

• Cloud Architecture: AWS Lambda, API Gateway, Amazon Rekognition, AWS S3 • Deployment & Automation: Docker, FastAPI, Git CI/CD, Redis, GitOps Agent • Edge Hardware & Media: NVIDIA Jetson, ARM Cortex, Raspberry Pi, FFmpeg, GStreamer	• AI-Driven Development: Cursor, Claude Code, Model Context Protocol (MCP), Agentic Workflows, Autonomous Software Systems • Core Languages: Python (Core Specialist), C, MATLAB
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CERTIFICATIONS

- Introduction to Model Context Protocol | Anthropic | July 2026 | [🔗](#)
- Introduction to Agent Skills | Anthropic | July 2026 | [🔗](#)
- Claude Code in Action | Anthropic | July 2026 | [🔗](#)
- AWS MLOps Training Certificate | Amazon Web Services | 2023 | [🔗](#)

SELECTED PUBLICATIONS

1. "Robust Reversible Data Hiding within Textures via Dual-Layer Encryption" – IEEE International Conference on Innovations in Information, Embedded and Communication Systems, March 2017 [🔗](#)
2. "An Effective Malayalam OCR with Loop, Line and Curve Features" – Elsevier Computer Networks and Security, March 2014 [🔗](#)
3. "A Level-Set Based Binarization and Segmentation for Scanned Malayalam Document Image Analysis" – IEEE International Conference on Computational Intelligence and Computing Research, December 2011 [🔗](#)